

EMPLOYMENT	NVIDIA Research	Apr 2026—Present
	<i>Senior Research Engineer</i> , Autonomous Systems and Physical AI Research Group - Develop foundation and edge models for Physical AI, including NVIDIA Alpamayo, advancing robust perception, reasoning, and action across diverse embodiments. - Build agentic research workflows that automate data curation, experimentation, and evaluation, accelerating model development and Physical AI dataset releases.	
	Waymo Perception	Jun 2024—Apr 2026
	<i>Software Engineer</i> , Semantics Foundation Team - Built and deployed onboard vision-language models (VLMs) to enhance self-driving safety and robustness in long-tail scenarios. - Led large-scale data mining, model evaluation, and system simulation to improve model reliability and real-world performance.	
EDUCATION	Cornell University	Aug 2019—Jun 2024
	<i>Ph.D.</i> , Computer Science	Advisor: Prof. Kilian Q. Weinberger
	Shanghai Jiao Tong University	Sep 2015—Jun 2019
	<i>B.S.</i> , Computer Science	Advisor: Prof. Cewu Lu
PUBLICATIONS	- Learning 3D Perception from Others' Predictions Jinsu Yoo, Zhenyang Feng, Tai-Yu Pan, Yihong Sun, Cheng Perng Phoo, Xiangyu Chen, Mark Campbell, Kilian Q Weinberger, Bharath Hariharan, Wei-Lun Chao ICLR 2025 [paper] - CornerPoint3D: Look at the Nearest Corner Instead of the Center Ruixiao Zhang*, Runwei Guan*, Xiangyu Chen, Adam Prugel-Bennett, Xiaohao Cai <i>arXiv preprint</i>, 2025 [paper] - In-Vitro Blood Purification using Tiny Pinch Holographic Optical Tweezers Based on Deep Learning Xiao Luo, Yu Ching Wong, Xiangyu Chen, Handong Tan, Weijia Wen Biosensors and Bioelectronics, 2025 [paper] - DiffuBox: Refining 3D Object Detection with Point Diffusion Xiangyu Chen*, Zhenzhen Liu*, Katie Z Luo*, Siddhartha Datta, Adhitya Polavaram, Yan Wang, Yurong You, Boyi Li, Marco Pavone, Wei-Lun Chao, Mark Campbell, Bharath Hariharan, Kilian Q. Weinberger NeurIPS 2024 [paper] - Reward Finetuning for Faster and More Accurate Unsupervised Object Discovery Katie Z Luo*, Zhenzhen Liu*, Xiangyu Chen*, Yurong You, Sagie Benaim, Cheng Perng Phoo, Mark Campbell, Wen Sun, Bharath Hariharan, Kilian Q Weinberger NeurIPS 2023 [paper] - Waymax: An Accelerated, Data-Driven Simulator for Large-Scale Autonomous Driving Research Cole Gulino*, Justin Fu*, Wenjie Luo*, George Tucker*, Eli Bronstein, Yiren Lu, Jean Harb, Xinlei Pan, Yan Wang, Xiangyu Chen, John D. Co-Reyes, Rishabh Agarwal, Rebecca Roelofs, Yao Lu, Nico Montali, Paul Mouglin, Zoey Yang, Brandyn White, Aleksandra Faust, Rowan McAllister, Dragomir Anguelov, Benjamin Sapp NeurIPS 2023 [paper]	

- Learning Iterative Neural Optimizers for Image Steganography
Xiangyu Chen*, Varsha Kishore*, Kilian Weinberger
[ICLR 2023 \[paper\]](#)
- Learning To Invert: Simple Adaptive Attacks for Gradient Inversion in Federated Learning
Ruihan Wu*, **Xiangyu Chen***, Chuan Guo, Kilian Weinberger
[UAI 2023 \[paper\]](#)
- Ithaca365: Dataset and Driving Perception under Repeated and Challenging Weather Conditions
Carlos A. Diaz-Ruiz, Youya Xia, Yurong You, Jose Nino, Junan Chen, Josephine Monica, **Xiangyu Chen**, Katie Z Luo, Yan Wang, Marc Emond, Wei-Lun Chao, Bharath Hariharan, Kilian Q. Weinberger, Mark Campbell
[CVPR 2022 \[paper\]](#)
- Hindsight is 20/20: Leveraging Past Traversals to Aid 3D Perception
Yurong You, Katie Luo, **Xiangyu Chen**, Junan Chen, Wei-Lun Chao, Wen Sun, Bharath Hariharan, Mark Campbell, Kilian Weinberger
[ICLR 2022 \[paper\]](#)
- Fixed Neural Network Steganography: Train the Images, not the Network
Varsha Kishore*, **Xiangyu Chen***, Yan Wang, Boyi Li, Kilian Weinberger
[ICLR 2022 \[paper\]](#)
- Low-Precision Reinforcement Learning: Running Soft Actor-Critic in Half Precision
Johan Bjorck, **Xiangyu Chen**, Christopher De Sa, Carla P. Gomes, Kilian Weinberger
[ICML 2021 \[paper\]](#)
- Train in Germany, Test in The USA: Making 3D Object Detectors Generalize
Yan Wang*, **Xiangyu Chen***, Yurong You, Li Erran Li, Bharath Hariharan, Mark Campbell, Kilian Weinberger, Wei-Lun Chao
[CVPR 2020 \[paper\]](#)
- Transferable Active Grasping and Real Embodied Dataset
Xiangyu Chen*, Zelin Ye*, Jiankai Sun, Yuda Fan, Fang Hu, Chenxi Wang, Cewu Lu
[ICRA 2020 \[paper\]](#)
- TendencyRL: Multi-Stage Discriminative Hints for Efficient Goal-Oriented Reverse Curriculum Learning
Chen Wang*, Junfeng Ding*, **Xiangyu Chen**, Zelin Ye, Jialu Wang, Ziruo Cai, Shixiang Gu, Cewu Lu
[IROS 2019 \[paper\]](#)
- Semantic Predictive Control for Explainable and Efficient Policy Learning
Xinlei Pan*, **Xiangyu Chen***, Qizhi Cai, John Canny, Fisher Yu
[ICRA 2019 \[paper\]](#)

HONORS
AND
AWARDS

ICM Outstanding Winner & Vilfredo Pareto Award (*Top 0.17%*)
National Scholarship of China (*Top 0.2%*)

2018
2018, 2017, 2016

PROFESSIONAL
SERVICES

Reviewer of CVPR (2019, 2022, 2026); ECCV (2022); ICLR (2025); ICRA (2019, 2020); NeurIPS (2023, 2024, 2026); RSS (2019, 2020); IEEE Transactions on Image Processing (2026)

Last Update: September 4, 2026